

```

#version 460
#extension GL_EXT_shader_16bit_storage : require
#extension GL_KHR_shader_subgroup_basic : require
#extension GL_KHR_shader_subgroup_arithmetic : require

layout(local_size_x = 64, local_size_y = 1, local_size_z = 1) in;

layout(set = 0, binding = 0) readonly buffer InputFP16_W64 {
    float16_t in_data16_w64[];
};
layout(set = 0, binding = 1) writeonly buffer OutputFP16_W64 {
    float16_t out_data16_w64[];
};

void main() {
    uint idx = gl_GlobalInvocationID.x;

    float16_t val = in_data16_w64[idx];
    float16_t res = val * val + float16_t(2.0);

    out_data16_w64[idx] = subgroupAdd(res);
}

```